SUPPORTING ADULT GUIDANCE

What is a digit worth?

Place Value · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This is likely the learner's first NEO lesson, so it is calm, visual and reading-light. In the main activity, the Digit Value Magnifier, the learner taps any digit in a place-value chart (up to the millions) and sees what it is worth, in figures and words. A second chart lets them slide one digit along the columns and watch its value grow ten times with each step. Six short questions follow, and an optional stretch asks why each column is worth ten times the next. Learners then practise moving between column names, words and figures, match large numbers to their words, and make numbers from four digits as a detective.

Driving Question: *Why is the 3 in 300 worth more than the 3 in 30?*

The mathematics, in plain terms

Our number system is positional and base ten: a digit's value is the digit multiplied by the place value of its column, and each column's place value is ten times the one to its right. Zero is a placeholder. Large numbers are grouped in threes (ones, thousands, millions) for reading. Decimals, and multiplying and dividing by 10, 100 and 1000, follow in Lessons 2 and 4.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — What is each digit worth?

In 4,073,250, what is the 7 worth? And what is the 0 after the 4 doing?

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
4	0	7	3	2	5	0

The 7 is in the ten-thousands column.

1. Write the number in the chart, one digit per column.	4 | 0 | 7 | 3 | 2 | 5 | 0
2. Read the heading above the 7.	ten thousands
3. Multiply the digit by its column.	7 × 10,000 = 70,000
4. Look at the 0 after the 4.	it holds the hundred-thousands column open

The 7 is worth 70,000, seventy thousand. Without the 0, the number would be 473,250.

Example 2 — Sliding a digit

What happens to the value of a 7 when it moves one column to the left?

7 in the hundreds column

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
				7	0	0

Slide it one column left

Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
			7	0	0	0

1. In the hundreds column it is worth	700
2. One column left, in the thousands column	7,000
3. Compare the two values.	7,000 = 700 × 10
4. Why?	a thousand is ten hundreds

Each step to the left makes a digit worth ten times as much, because each column holds ten of the column to its right.

Picture, then reason, then words

Learners see the value of each digit before naming any rule, and the slider shows the tenfold relationship as it happens. The stretch then explains it: ten of one column make one of the next. Reading and writing large numbers comes after, because grouping in threes only makes sense once the columns do.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Digit Value Magnifier: tap digits, slide a digit, six questions. Stretch: why ten times?	~15 min
Movement	Column hop: column names, words and figures.	~7 min
Reflection	Read it, write it: large numbers matched to their words.	~8 min
Creativity	Digit detective: four digits, five cases.	~10 min

Rest	Ten symbols are enough for any whole number.	~2 min
Nutrition	Intellectual: a sense of size.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Giving the digit instead of its value (7 instead of 70,000).
Ask: “Which column is it in? So what is it worth?”

Dropping a placeholder zero (writing 4,050 as 450).
Ask: “How many columns does four thousand need? Which one is empty?”

Reading a number digit by digit instead of in groups of three.
Ask: “Can you split it into groups of three from the right first?”

Thinking the number with “bigger digits” is larger (2,099 against 2,109).
Ask: “Compare from the left. Where is the first difference?”

Saying “add a zero” to multiply by 10.
Ask: “What happens to each digit? Which column does it move to?” The zero fills the empty ones column.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: place value up to the thousands. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says the 7 in 4,073,250 is worth 7,000. The aim is a question that lets them find the column themselves.

- “Can you put each digit in the chart?”

- “Count the columns from the right: what is the fifth one called?”
- “So what is the 7 worth?”
- “What would a 7 be worth one column further left?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners write commas in large numbers?
Either is fine. The lesson accepts 70,000, 70 000 and 70000, and number words. Commas help with reading in groups of three.

Why not teach “add a zero” for × 10?
It breaks with decimals (2.5 × 10 is not 2.50). Lesson 2 shows the digits moving instead.

Is the stretch required?
No. The secure core is stating the value of any digit and reading and writing large numbers.

What comes next?
Lesson 2: ten times bigger, ten times smaller.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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